Final Project Report

Group 56

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Part 1: Force-Displacement Curve Fitting and Model Selection

1. Introduction

The purpose part 1 is to create a smooth, differentiable function F(x) given a set of isolator force data. The function must properly represent the behavior of the vibration isolation mount over the given displacement range ($0 \leq x \leq 0.020$ m). During this part, the group fit the data with a third order polynomial and a physics-guided nonlinear polynomial ($F(x) = k_1x + k_2\tan(k_3x)$). The group used MATLAB to solve the unknown parameters of the nonlinear polynomial, plot the nonlinear function against the third order polynomial and raw data points, plot the residual error for each model, and create a table with the respective RMSE and R^2 values.

2. Results

a. Task 7 Plot

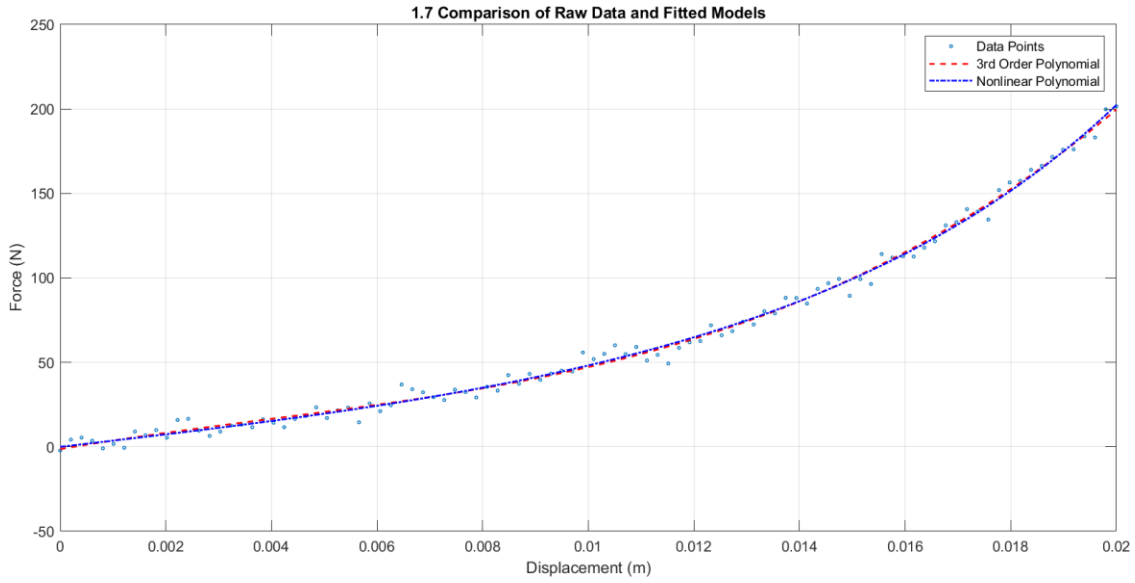


Figure 1. Curve Fitting Comparison

b. Task 8 Plot

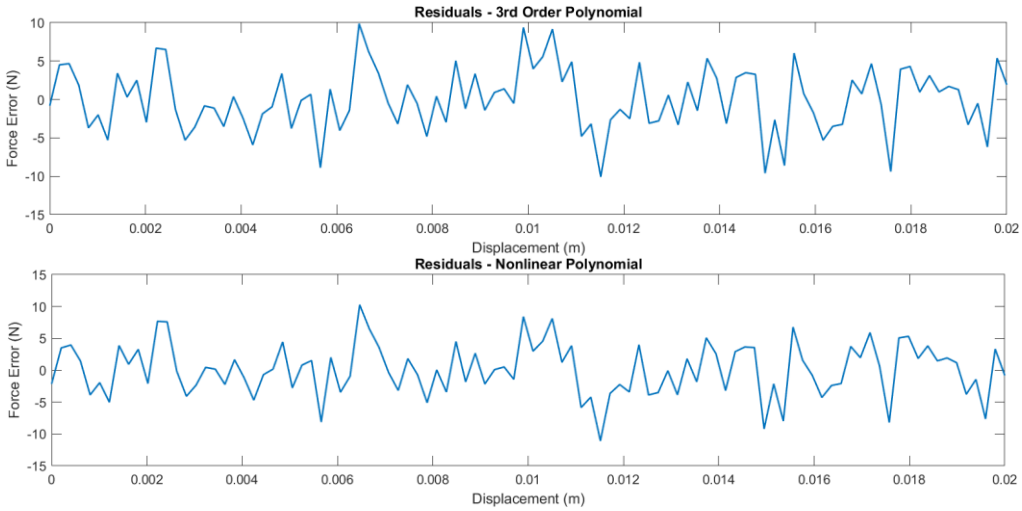


Figure 2. Curve Fitting Residuals

c. Task 8 Table

Table 1. Model Comparison Table

Model	RMSE	R^2
3rd Order Polynomial	4.1432	0.99429
Nonlinear	4.1084	0.99438

3. Discussion

1. The nonlinear model better represents the behavior of the isolator. The RMSE value shows the average magnitude of error is less than the 3rd order polynomial by 0.0348. Additionally, the R^2 value shows that the nonlinear model better predicts the variation of the data compared to the 3rd order polynomial.
2. The nonlinear model is more likely to produce a more stable derivative in part 2. Using high order polynomials, such as a 3rd order or higher, can cause Runge's phenomenon at the start and end of the data. The nonlinear model is also "physics-guided" meaning it would better predict an isolator's motion as opposed to the cubic function which does not care about physics.
3. Smoothness is especially important before numerical differentiation because it amplifies error. In terms of the problem, the stiffness of the isolator, determined from numerical differentiation, should change gradually as it compresses. A smooth model will ensure the derivative changes gradually and continuously.
4. Measurement noise and sparse data will cause the nonlinear model to be even more accurate than the 3rd order polynomial. Higher order polynomials are sensitive to noise because they try to pass through every point (higher R^2) but make no physical sense. Sparse data causes models to be lower order polynomials which lack flexibility for curve fitting.

Part 2: Local Stiffness from Numerical Differentiation

4. Introduction

The goal for Part 2 is to estimate the local stiffness of the vibration isolator from the fitted force-displacement model. Since the nonlinear model was picked for Part 1, the function is $f(x) = k_1 x + k_2 \tan(k_3 x)$. The local stiffness was found by numerically differentiating the functions $k_{local}(x) = \frac{df}{dx}$. The calculations were repeated for N = 10, 20, 40, 80, 160. The analytical derivative of the fitted model was used for the reference solution when calculating error.

5. Results

a. Task 3 Plot

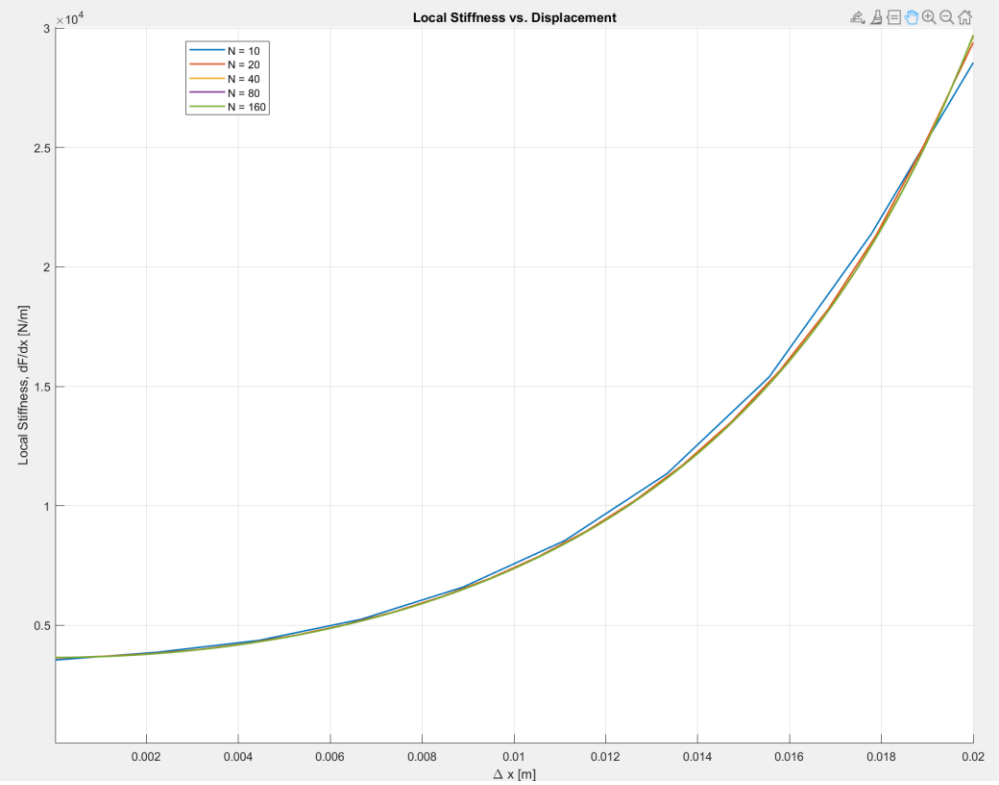


Figure 3. Local Stiffness vs. Displacement

This plot shows the Local Stiffness versus Displacement for different grid sizes. As N increases, the curves become smoother and get closer together. The Local Stiffness also increases with displacement, indicating that the isolator is stiffer the more it is compressed.

b. Task 6 Plot

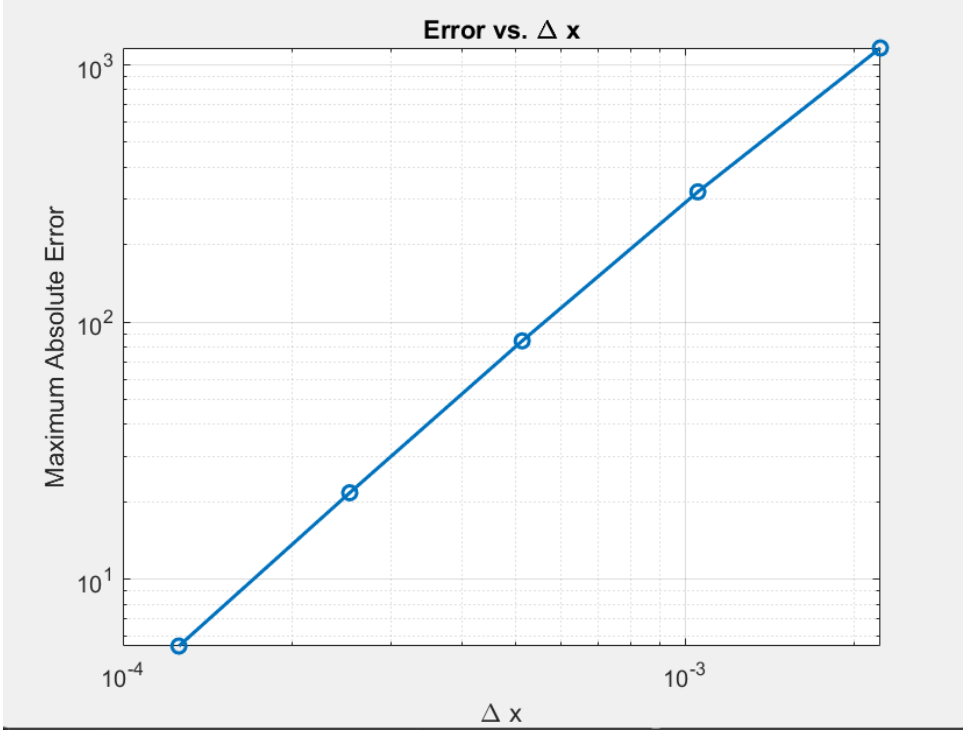


Figure 4. Maximum Error vs. Delta_x

This plot shows the maximum error versus the delta_x on a log-log scale. The error increases along with delta_x, showing that the numerical derivative converges toward the analytical derivative.

c. Written Interpretation

The Local Stiffness increases as the displacement increases, which means that the isolator behaves nonlinearly. As N increases, the numerical stiffness curves become smoother and approach the same shape. Also, the error plot shows that the maximum error decreases as delta_x decreases. The observed order of accuracy is about 2, which matches what we expect to see for a second-order finite-difference method.

6. Discussion

1. It is most sensitive towards the far right, where x = 0.020. This is due to the force curve changing at the fastest rate there, so small errors in the spacing can cause bigger changes for the derivative. Also, since we can't use central differences at the ends, it will be more sensitive.
2. Yes, the stiffness curves become smoother as N increases. The wider grids show a more noticeable difference, whilst the finer grids show a smaller difference. This proves that the derivative estimate is converging as N increases.
3. Yes, the observed order is 1.87, which is close to the second-order accuracy of 2. Since the method is based on second-order differences in the interior of the graph and second-order one-sided formulas at the ends of the graph, our result is expected to be near 2.

4. Boundary points are more delicate because the central difference formula can't be used at boundary points. At the ends, there are no central data points outside of the interval, so we have to use the one-sided formula. These formulas are generally more sensitive, especially when the slope is changing fast.

Part 3: Strain Energy from Numerical Integration

7. Introduction

The goal of Part 3 is to estimate the strain energy stored in the vibration isolator. It does this by numerically integrating the fitted force-displacement model from Part 1. Since the nonlinear model was selected in Part 1, the force function we used was $f(x) = k_1 x + k_2 \tan(k_3 x)$. The strain energy was calculated over the interval $0 \leq x \leq 0.020$ using the integral $U = \int_0^{0.020} f(x)dx$. Both the Trapezoidal rule and the composite Simpson's 1/3 were used for N = 10, 20, 40, 80, and 160 intervals. The graph of the fitted force model was used to find the reference solution so that numerical error and convergence behavior can be looked at.

Results

a. Task 2 Table

N	Delta_x	U_trap	U_simp
10	0.002	1.2968	1.2882
20	0.001	1.2903	1.2881
40	0.0005	1.2887	1.2881
80	0.00025	1.2883	1.2881
160	0.000125	1.2882	1.2881

Table 2. Strain Energy Results Table

This table shows the results for both the Trapezoidal rule and Simpson's 1/3 rule for differing values of N and Delta_x. As the grid gets refined, or N increases and Delta_x decreases, bot methods approach the same value; however, Simpson's 1/3 rule is closer at every size.

b. Task 6 Plot

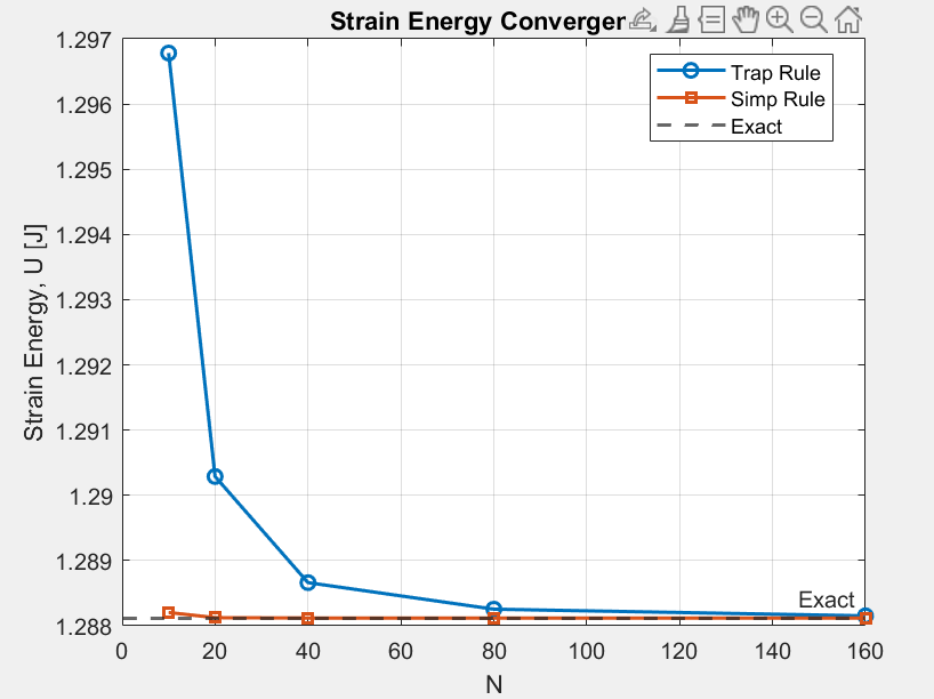


Figure 5. Strain Energy Convergence

This plot shows the estimates from both the Trapezoidal rule and Simpson's 1/3 rule. Both methods approach the same value as the intervals increase, but Simpson's rule approaches it much faster.

c. Task 6 Plot

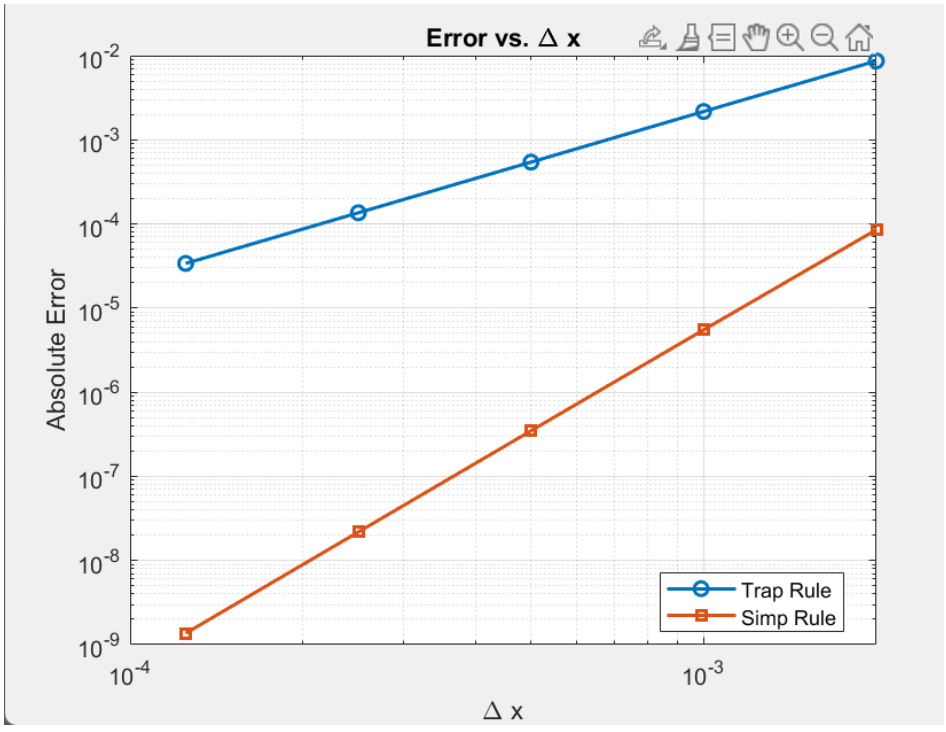


Figure 6. Integration Error Plot

This plot shows the integration error versus the grid spacing on a log-log scale. The Trapezoidal error shows second-order convergence, while Simpson's 1/3 rule shows more of a fourth-order convergence. This means that the Simpson's error decreases faster as delta_x increases.

5. Discussion

1. Yes, both methods converge towards the same strain energy value of 1.2881186104. As the N value increases, both estimates get closer to this value; however, the Simpson's 1/3 rule gets closer at a much faster rate than the Trapezoidal rule. This makes sense because they are both approximating the same definite integral, so they show an estimate of the same value.
2. Simpson's 1/3 rule converges faster because Simpson's rule has a higher order of accuracy than the Trapezoidal rule. So, in comparison to the Trapezoidal rule, Simpson's will give a much more accurate estimate of the force curve.
3. One big error with a fine-grid numerical reference instead of an exact analytical value is that the fine-grid numerical reference is only an approximation. Even if we make the grid super small, there will still be some level of error. The analytical value is better because it gives the exact strain energy for the force curve, not an approximation.
4. The observed order of accuracy is the log-log error versus delta_x. For the Trapezoidal rule, the slope is 2, which means that the error decreases by a factor of 4 when delta_x is cut in half. The Simpson's 1/3 rule has a slope of 3.98, which means that the error decreases by about a factor of 16 when delta_x is cut in half. Simpson's rule improves much faster, so the larger the slope, the faster the methods of error shrink as N increases.
5. The smoother the fitted curve is, the better both methods will be. With a smoother curve, the methods have an easier time approximating the values accurately. Simpson's 1/3 rule will especially benefit because it uses parabolic behavior to approximate the curve. This means if there are more sharp jumps or jaggedness to the curve, the approximations would be less accurate.

Part 4: Transient Response of a Mass–Spring–Damper System

6. Introduction

In part 4 of the project, we modeled the transient response of the isolation system was created using a single degree of freedom mass spring dampener system equation that follows:

$$f(t) = m\frac{d^2x}{dt^2} + c\frac{dx}{dt} + kx$$

Where:

- m = 12 kg
- c = 95 Ns/m
- k = 4200 N/m.

Furthermore, the default forcing function that was applied followed:

$$f(t) = 120\sin(2\pi(2.5)t) + 60\sin(2\pi(7.0)t)$$

With the initial conditions:

- x(0) = 0
- $\dot{x}(0) = 0$

This was followed up with the system being solved over 0 <= t <= 10, where an analysis of solver step sizes, tolerance sensitivity, and a 5000-point uniform resampling. In order to solve the second order ODE in MATLAB, the damper system equation was rearrange and rewritten in the form of first order equations which can be seen in the following figure.

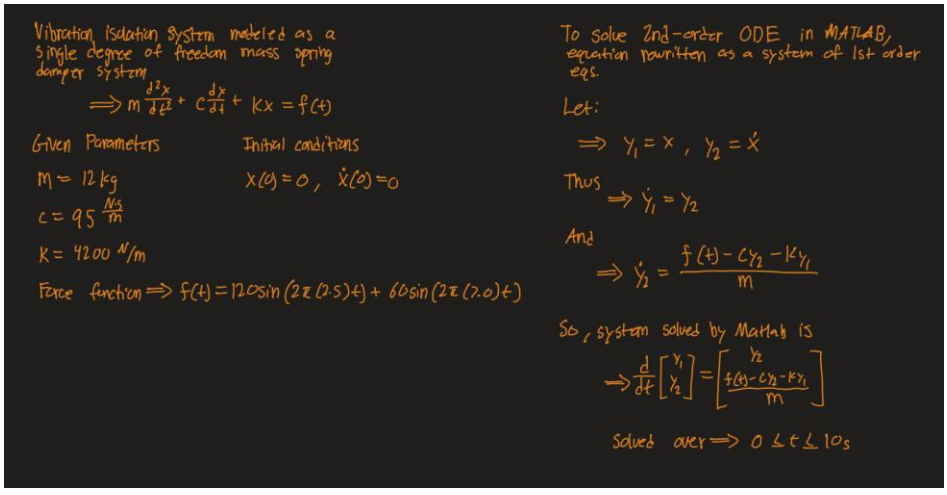


Figure 7. Governing Equation Conversion

The 1st order system was solved by MATLAB's built in ode45 solver.

7. Results

a. Task 3 Plot

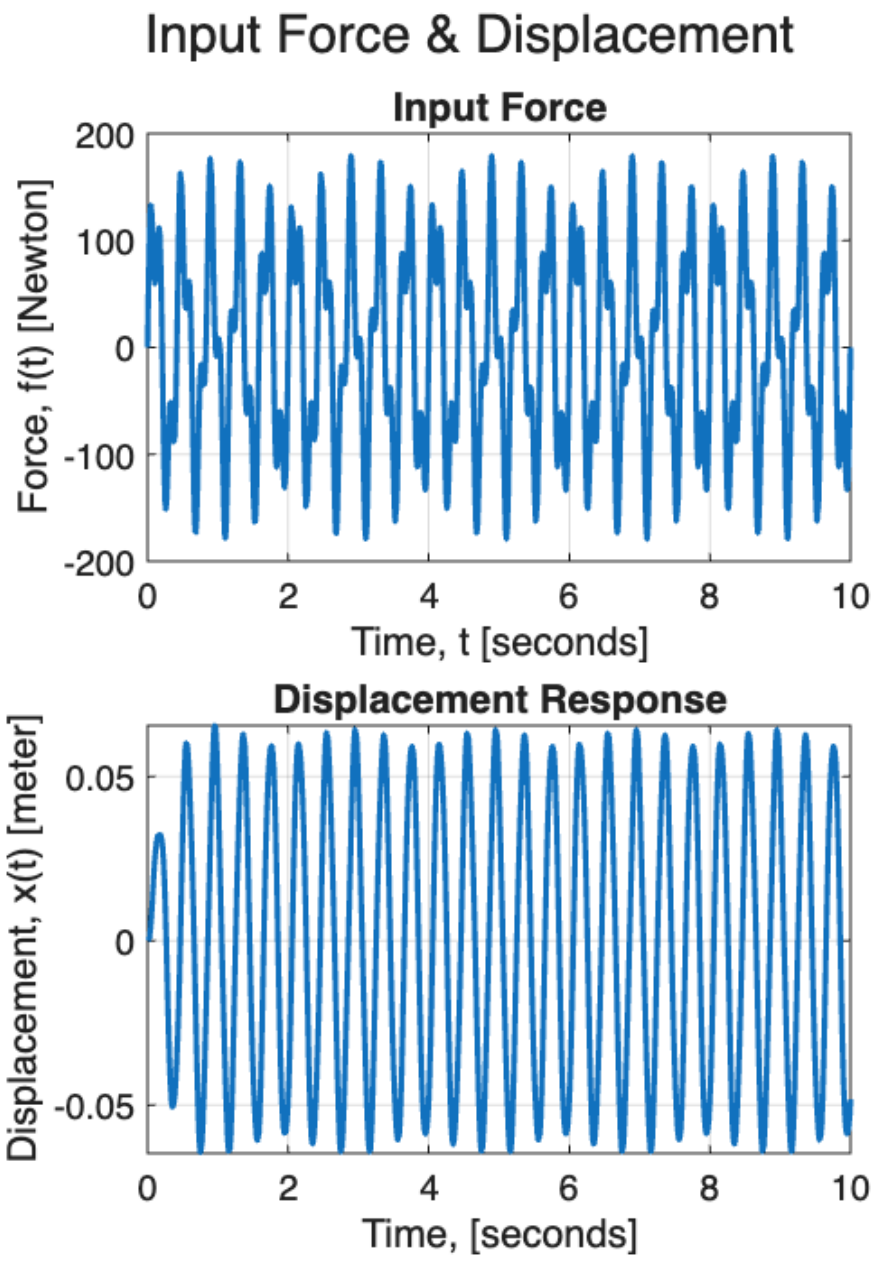


Figure 7. Input Force and Displacement Response

a. Task 4 Plot

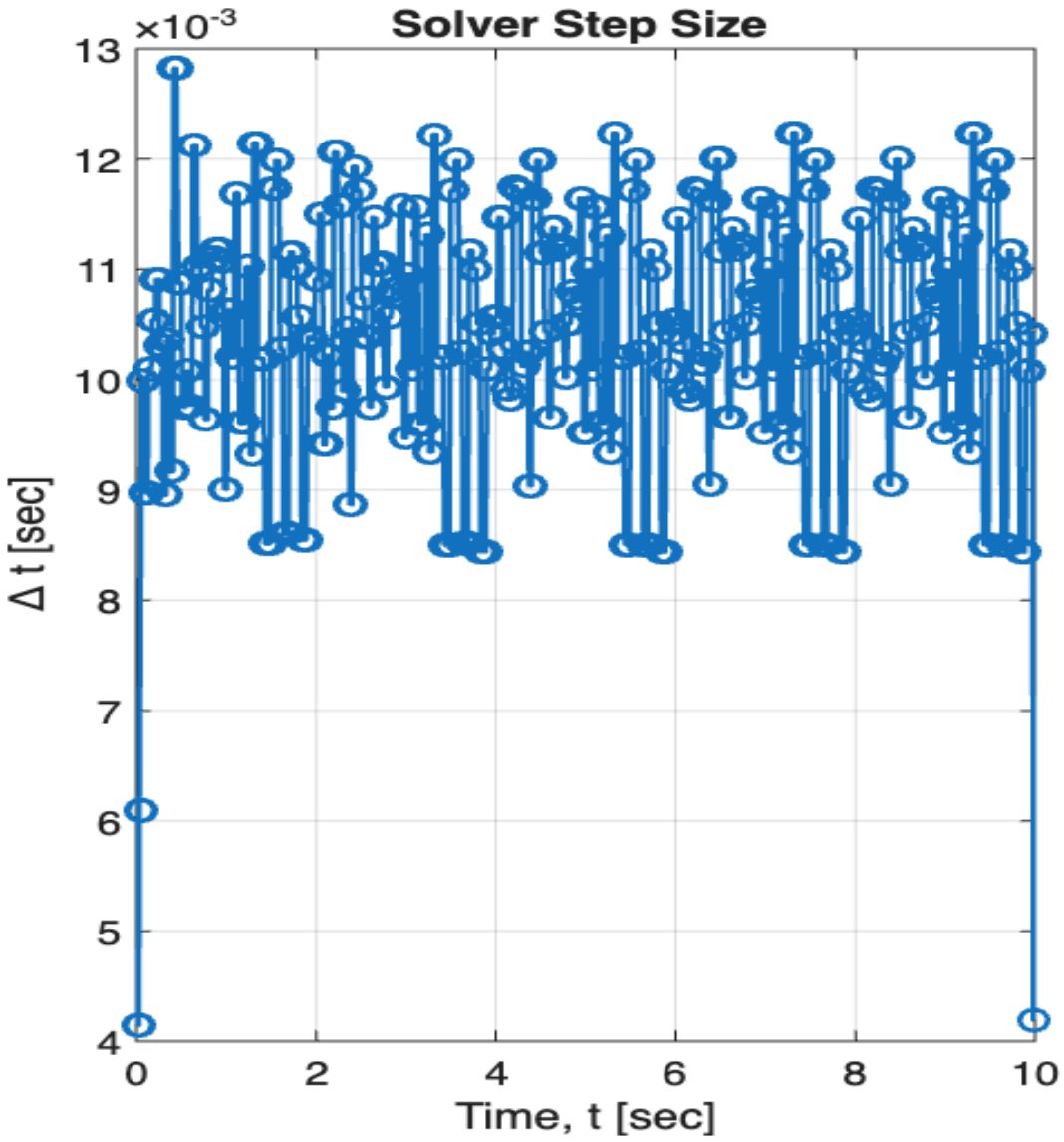


Figure 8. Solver Step Size

b. Task 5 Plot

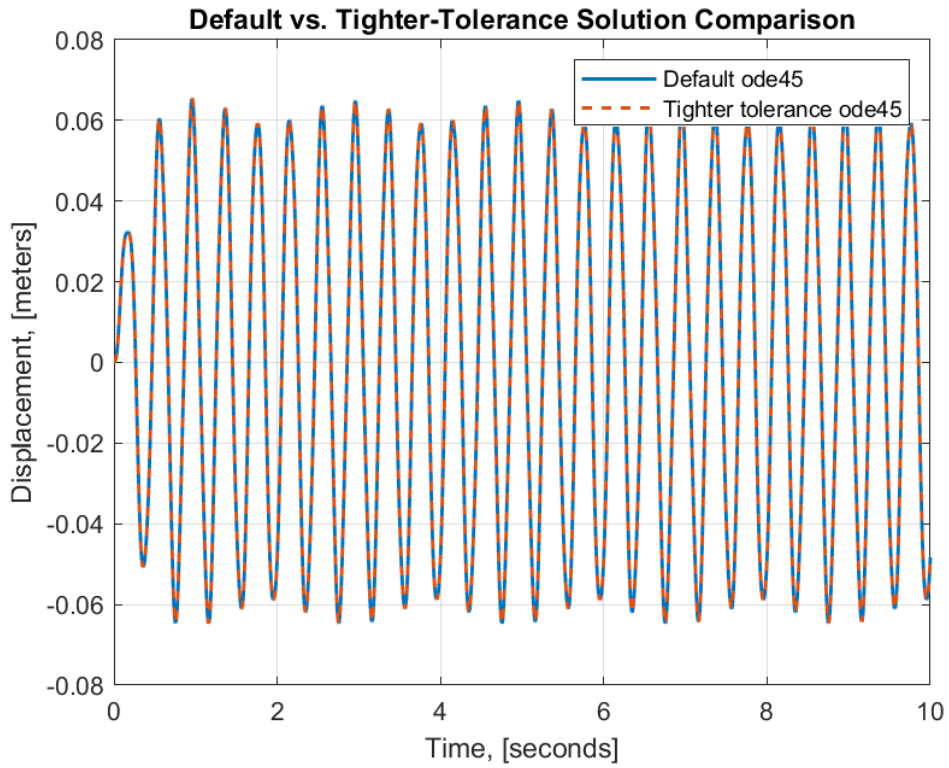


Figure 9. Comparison of Default and Tighter Tolerance

b. Task 5 Plot

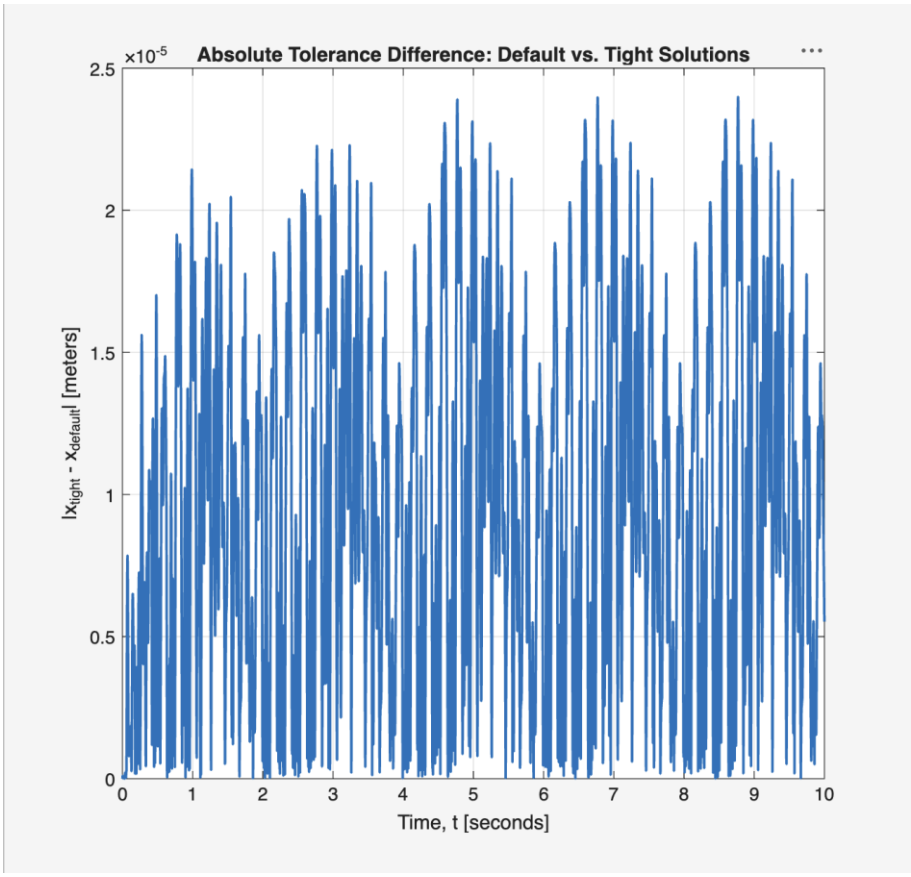


Figure 10. Difference Between Default and Tight Solutions

c. Task 6 Plot

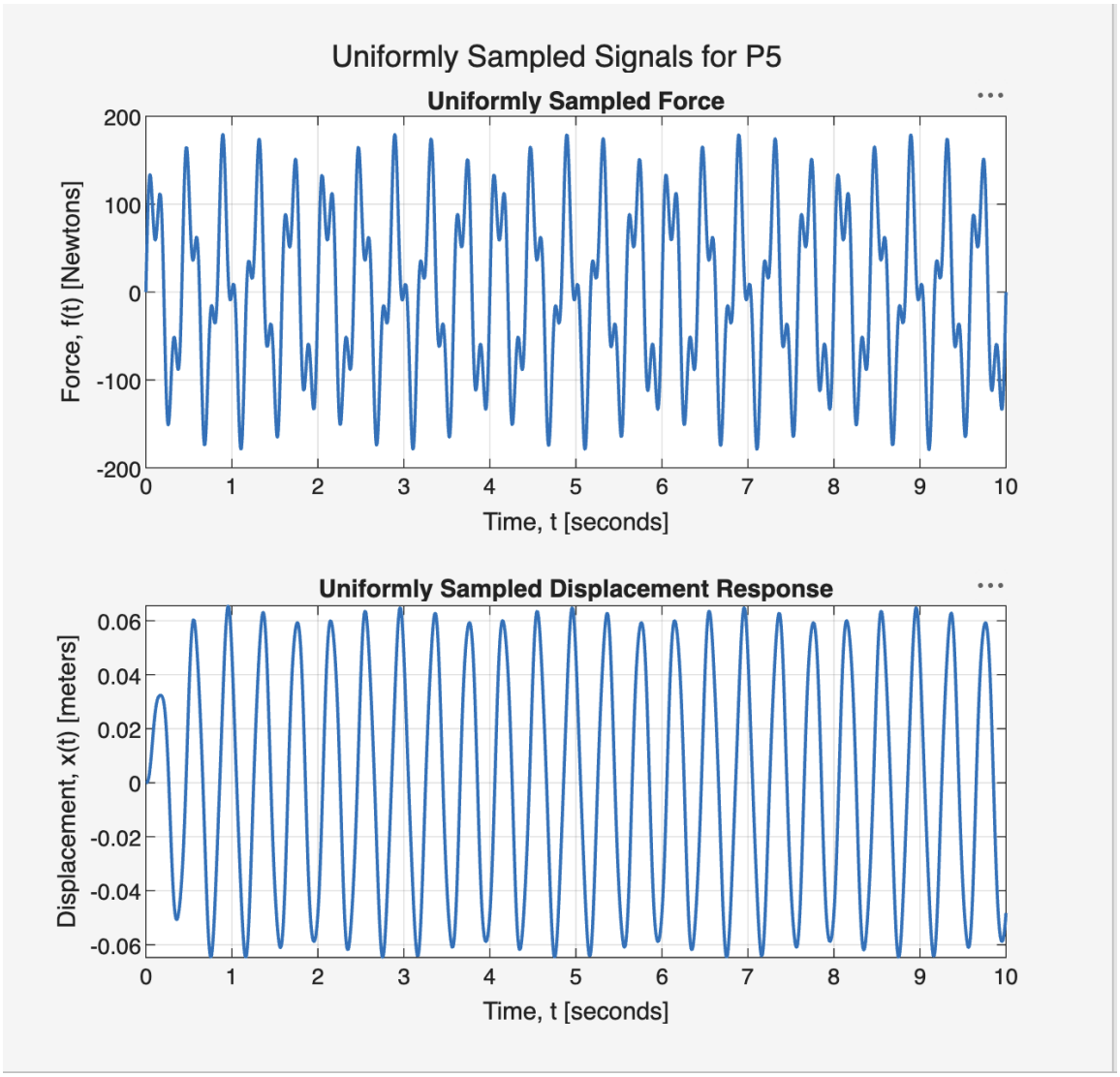


Figure 11. Uniformly Spaced Time Grid

8. Discussion

1. MATLAB's built in ODE solver utilizes the Runge-Kutta set of numerical methods. The function ode45 was required to work through this project because the second order mass spring damper equation had to be rewritten as a first order initial value problem.
2. Error order of the ODE method describes how fast numerical error decreases once the time step is refined. In practical terms, it can be understood as a higher order method typically can produce accurate results while not being dependent on a very small, fixed time step everywhere.
3. The solver used adaptive time steps, and this can be seen when computing $\Delta t_i = t_{i+1} - t_i$, where the smallest step size was 4.149057e-3 seconds, the mean step size was 1.02886e-2 seconds, and the largest step size was 1.281981e-2 seconds. Due to the differing values, the solver needed to change its time step throughout the simulation.

4. You must think carefully about time spacing because FFt analysis runs on the assumption of uniformly spaced data. Thus, the raw ode45 output used adaptive, nonuniform time steps, and the response then had to be interpolated onto a uniform 5000 point grid before moving on to part 5. This output gave $\Delta t = 2.00040 \times 10^{-3}$.

Part 5: Fourier Analysis, Sampling, and Signal Reconstruction

8. Introduction

The frequency content of the vibration response from part 4 was analyzed using the Fast Fourier Transform (FFT). FFT analysis has the constraint that time data must be uniformly spaced, thus the 5000 point uniformly sampled displacement response from part 4 was utilized as the master signal in this instance. The point of this was to avoid needing to apply the FFt directly to the ode45 output, which used adaptive nonuniform time steps.

To test what effects come from the sampling resolution, the master displacement signal was down sampled into three separate cases over the same time interval where $0 \leq t \leq 10$ seconds.

Cases:

- N = 50
- N = 250
- N = 1000

For each case, the FFT identified the dominant frequency content of the displacement response. Then the frequency resolution, sampling frequency, and nyquist frequency were also determined to justify if the results of each sampled signal could be used to accurately represent the important vibration frequencies. Furthermore, each sampled signal was eventually reconstructed and plotted in comparison with the initial 5000 point master signal. The comparison was a means to evaluate the reconstruction quality as the sampling resolution got larger. Finally, because the given applied force had sinusoidal inputs at 2.5 Hz and 7.0 Hz, the calculate FFT results in theory should show frequency content around those values for the system to be considered reasonable.

9. Results

a.

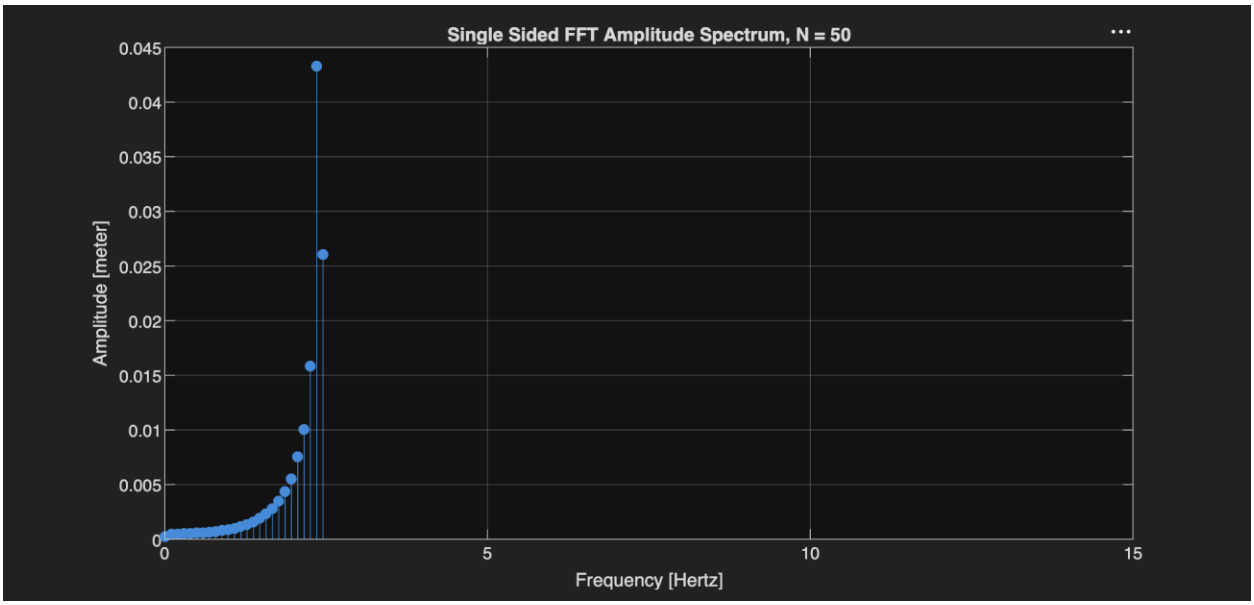


Figure 12. FFT Stem Plot: N=50

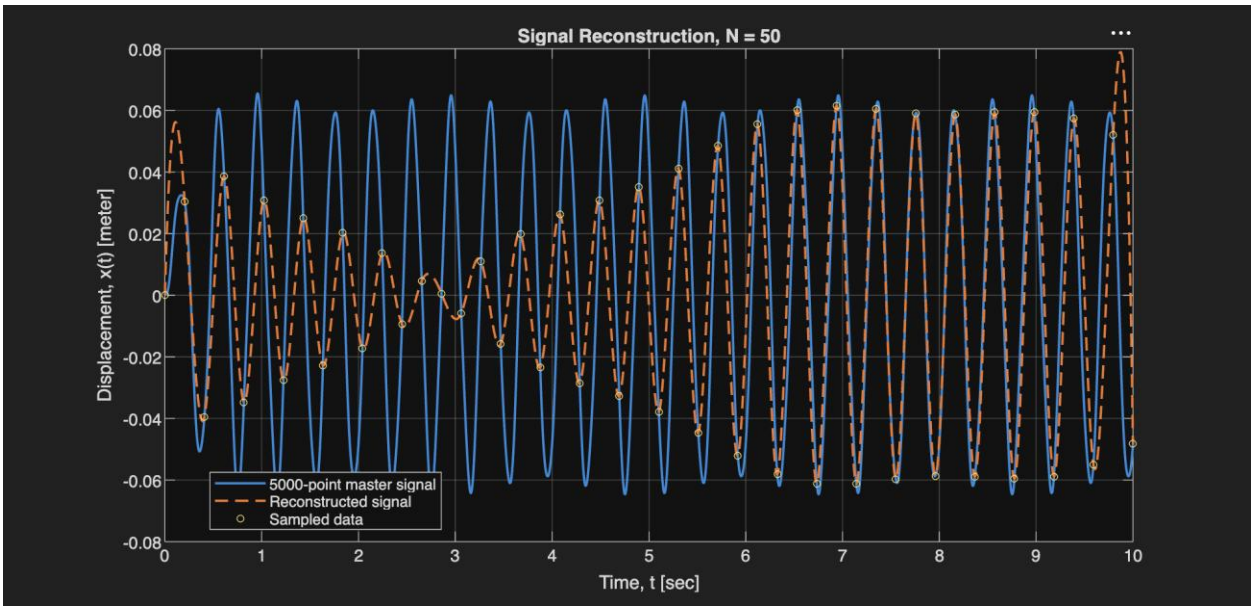


Figure 13. Signal Reconstruction Plot

b.

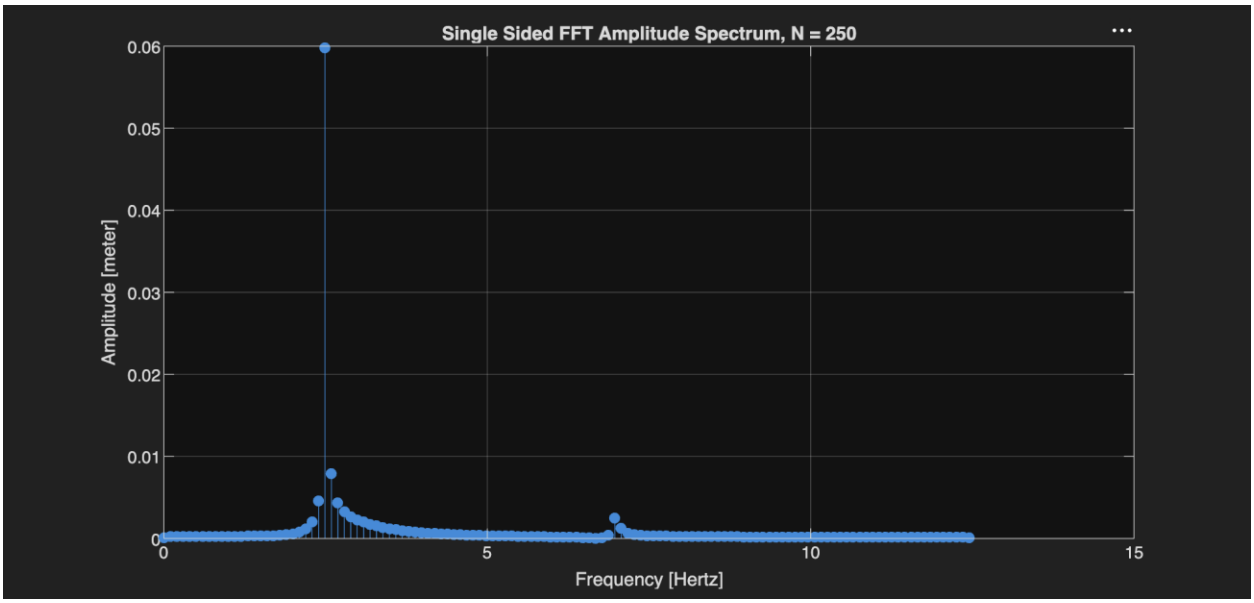


Figure 14. FFT Stem Plot: N=250

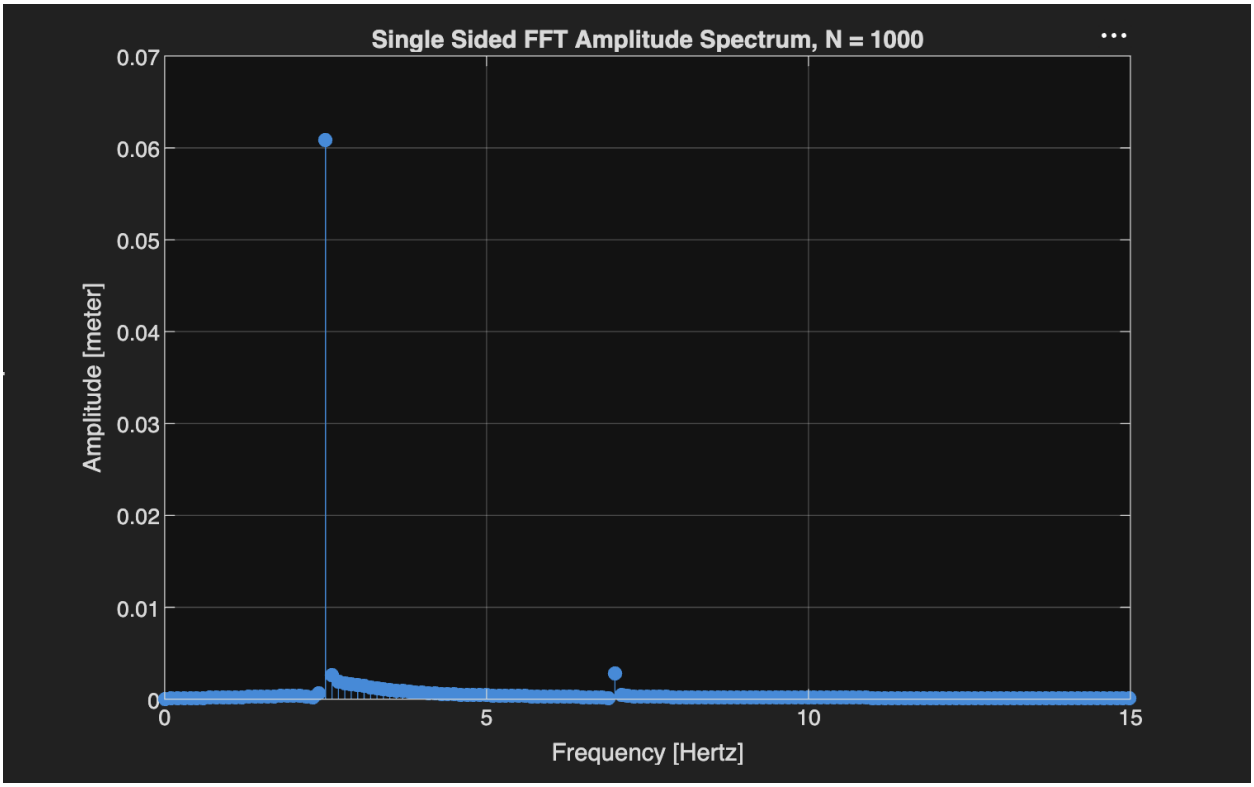


Figure 15. FFT Stem Plot: N =1000

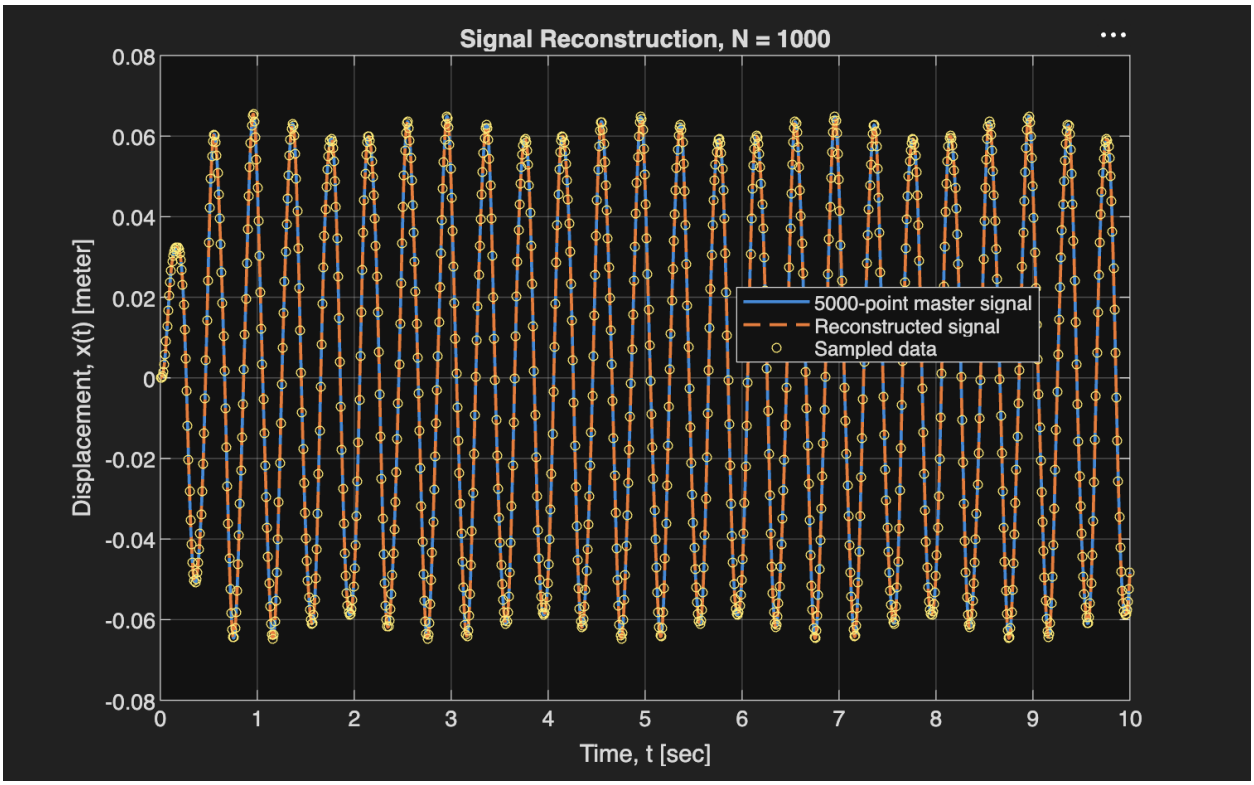


Figure 16. Signal Reconstruction: N=1000

9. Discussion

1. In the event of a high sampling resolution, the frequency estimate would be closer to the expected value, and it would allow higher frequency content to be captured at a more accurate level. When N = 50, we saw the dominant frequencies(Top freq. Peaks) in the range of 2.35-2.45 Hz, and subsequently for N = 250, 1000, the dominate frequency shifted closer to the expected 2.5 Hz input.
2. N = 1000 case provided the most accurate reconstruction as it had the smallest max reconstruction error of . The 250 and 50 cases were and .
3. Because when the sampling frequency is too low, it can't capture the true signal frequencies. When N = 50, the Nyquist frequency was determined to be 2.45 Hz, thus the signal couldn't actually be represented accurately as 2.45 Hz is below both of the forcing components of 2.5 and 7.0 Hz.
4. Once you are near 2.5 Hz the isolator has its strongest response, which it should be noted is reasonably close to the systems natural frequency of around 2.98 Hz. The natural frequency isn't given but can be calculated using the given parameters where , and plugging in k and m gives 18.708 rad/s. You then just need to convert to Hz using . When N =1000, our FFT provides a dominate peak at 2.4975 Hz with a corresponding amplitude of 0.0608 m. Then when you compare this to 7.0 Hz component, it appears much smaller at 6.9930 Hz paired with an amplitude of 0.0028 m.
5. In a real experiment, arbitrarily high sampling rates may be bound by constraints such as the system used for the data acquisition may have a maximum sampling rate. Another

practical limit that could cap sampling rates is the amount of noise that is also collected at high frequencies that would make the FFT results unclear, and the only way to avoid this would be to filter out the noise by means of filtration(low-pass, band-pass) but in doing so you would risk removing actual vibration content if the noise overlaps with the frequency in questions, thus affect results negatively. Overall, when the sampling rate reaches a high enough point where the capture of important frequencies is possible, it would most likely not benefit you to increase the rate any more, as this may not improve the results by a significant margin.

Team Contribution Statement

10. Jonathan Herskovits

a. Contribution 1

Completed Part 1: Force-Displacement Curve Fitting and Model Selection

b. Contribution 2

Created the Report Outline

c. Contribution 3

Setup Group Meeting

11. Justin Lewis

a. Contribution 1

Completed Part 4: Transient Response of a Mass–Spring–Damper System

b. Contribution 2

Completed Part 5: Fourier Analysis, Sampling, and Signal Reconstruction

12. Will Dalton

a. Contribution 1

Completed Part 2: Local Stiffness from Numerical Differentiation

b. Contribution 2

Completed Part 3: Strain Energy from Numerical Integration

Academic Integrity and AI Use Statement

Our team used AI mainly to help organize and direct the workflow of our report. We wanted to make sure that our report looked neat and had proper grammar, so we prompted the AI to review it and give us some suggestions on things to change, but we made sure that we did not let the AI generate anything, as that would go against academic integrity. The AI that we used was mainly ChatGPT, and whenever we prompted the AI system, we always included a statement at the end to make sure it would not generate things like code, answers, or grammatical rewrites for us.

This statement confirms that AI was not used to generate the report or code. Our team can fully understand and explain all the work submitted with this report.